

Scalability & Future Plan

Date	7 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a fixed dataset	Cannot learn new crop data automatically	High
2	Supports only crop recommendation	No fertilizer or disease prediction	Medium
3	Internet required for hosted version	Offline access is limited	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users	Deploy on cloud with auto-scaling
2	Data Storage	CSV/SQLite database	Upgrade to MySQL or PostgreSQL
3	Performance	Suitable for small datasets	Optimize model and use caching
4	Security	Basic input validation	Add user authentication and HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Improve crop recommendation accuracy	1 Month	Better prediction results
Phase 2	Add fertilizer recommendation	2 Months	Helps farmers improve yield
Phase 3	Add plant disease detection using images	3 Months	Early disease identification
Phase 4	Develop Android mobile application with weather integration	4–6 Months	Easy access for farmers anywhere